TRENDSTORE

Github Repo : https://github.com/wikkics/Trend

Docker Hub Image : wikkics/trend-app:latest

Jenkins Master URL : http://44.211.48.127:8080

EKS Cluster Name : trend-app-cluster

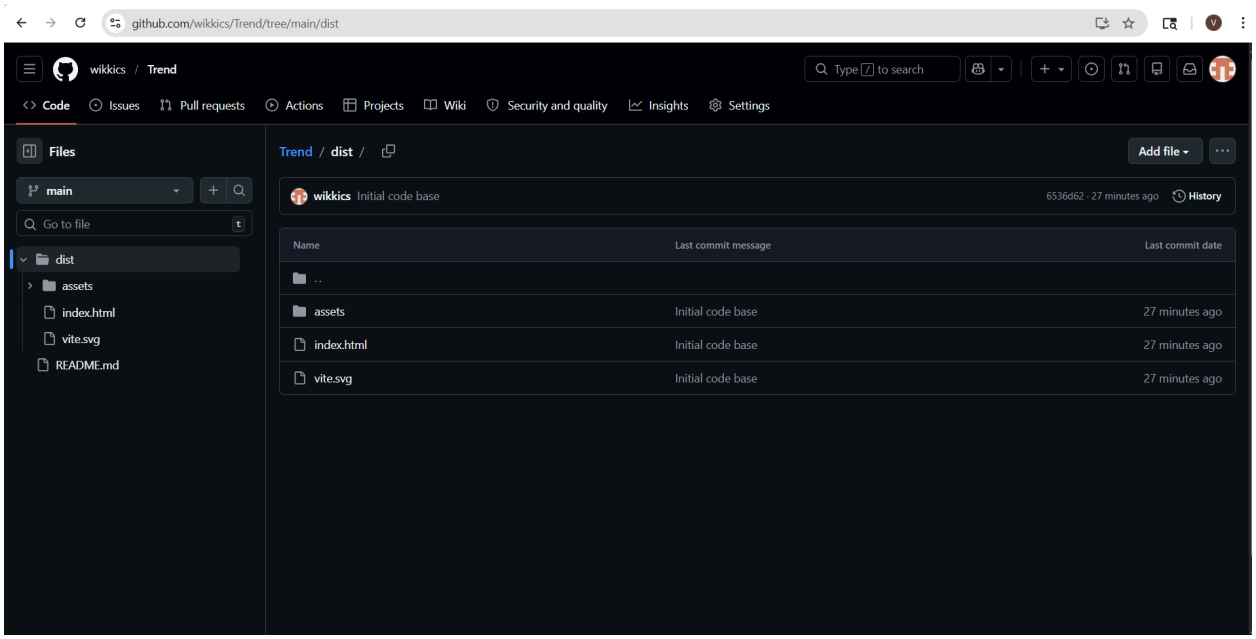
AWS Region : us-east-1

Load Balancer DNS
a8e97c6c648284fe28ea88911586e1bc-1412310142.us-east-1.elb.amazonaws.com

Production App URL
http://a8e97c6c648284fe28ea88911586e1bc-1412310142.us-east-1.elb.amazonaws.com

Source Code Management (SCM)

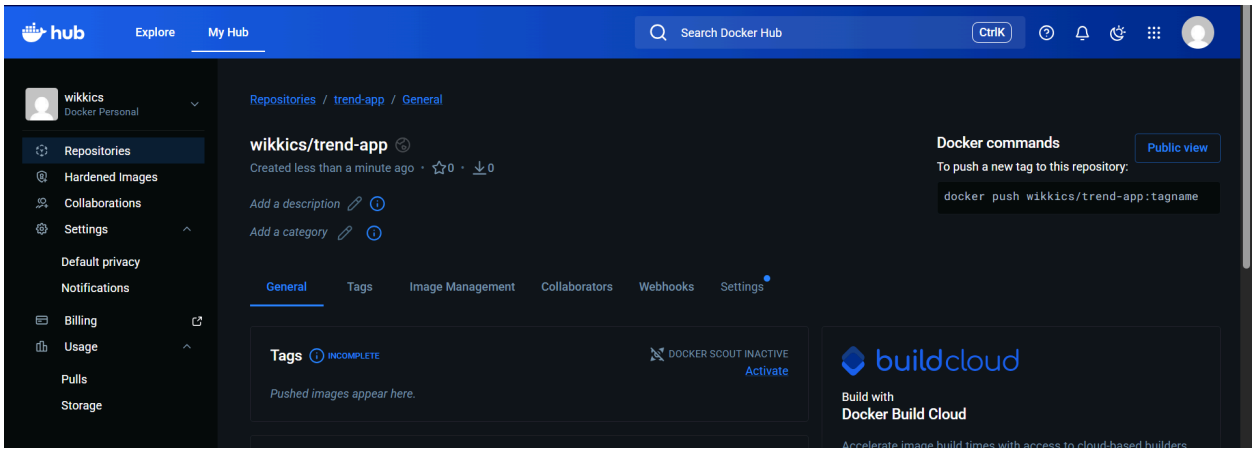
- Established a centralized Git repository to manage the React-based frontend code.
- Pushed the initial codebase, including the production dist folder, to GitHub for CI/CD integration.

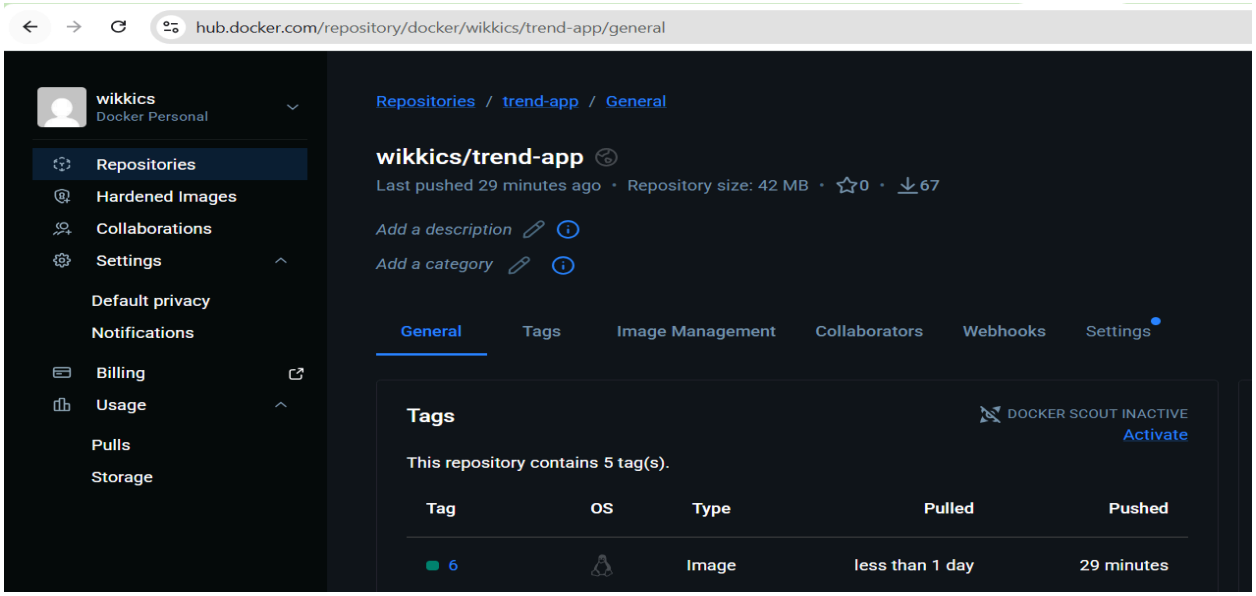


Containerization & Registry

- **Action:** Developed a multi-stage `Dockerfile` to build the app with Node.js and serve it via Nginx.
- **Result:** Built and tagged version-controlled images which were successfully pushed to Docker Hub.

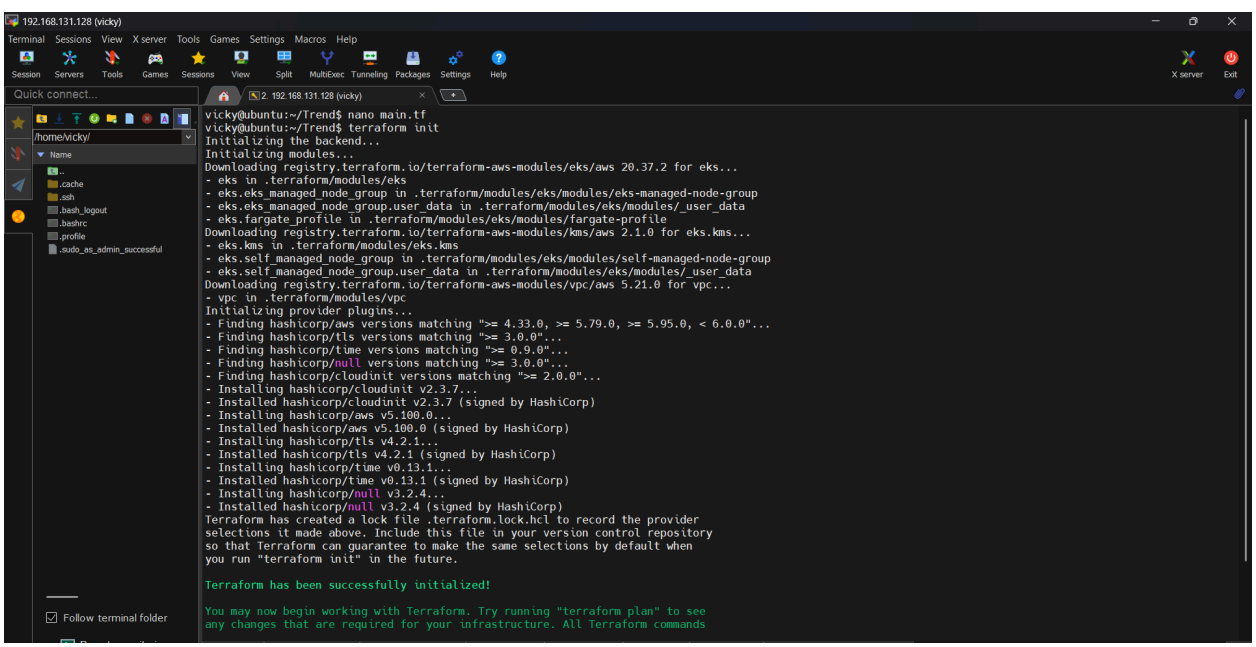
```
vicky@ubuntu:~/Trend$ pwd
/home/vicky/Trend
vicky@ubuntu:~/Trend$ nano DockerFile
vicky@ubuntu:~/Trend$ nano Dockerfile
vicky@ubuntu:~/Trend$ cat Dockerfile
# Stage 1: Build the React Application
FROM node:18-alpine AS builder
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .
RUN npm run build

# Stage 2: Serve the Application with Nginx
FROM nginx:alpine
# Update Nginx to listen on port 3000 instead of default 80
RUN sed -i 's/listen  *80;/listen 3000;/g' /etc/nginx/conf.d/default.conf
COPY --from=builder /app/build /usr/share/nginx/html
EXPOSE 3000
CMD ["nginx", "-g", "daemon off;"]
vicky@ubuntu:~/Trend$
```





Infrastructure as Code (IaC)

- Deployed a custom VPC, EKS Cluster, and an Ubuntu-based Jenkins instance using AWS provider configurations.
- A custom VPC with public/private subnets and a NAT Gateway.
- An EKS Cluster (`trend-app-cluster`) with a managed `t3.micro` worker node group
- A standalone Ubuntu EC2 server for Jenkins, secured via custom security groups and an attached SSH key pair.



```
vicky@ubuntu:~/Trend$ cat main.tf
provider "aws" {
  region = "us-east-1"
}

# 1. VPC setup
module "vpc" {
  source  = "terraform-aws-modules/vpc/aws"
  version = "~> 5.0"

  name = "trend-app-vpc"
  cidr = "10.0.0.0/16"

  azs             = ["us-east-1a", "us-east-1b"]
  private_subnets = ["10.0.1.0/24", "10.0.2.0/24"]
  public_subnets  = ["10.0.101.0/24", "10.0.102.0/24"]

  enable_nat_gateway = true
  single_nat_gateway = true
}

# 2. EKS Cluster
module "eks" {
  source  = "terraform-aws-modules/eks/aws"
  version = "~> 20.0"

  cluster_name    = "trend-app-cluster"
  cluster_version = "1.29"

  vpc_id     = module.vpc.vpc_id
  subnet_ids = module.vpc.private_subnets

  cluster_endpoint_public_access = true

  eks_managed_node_groups = {
    app_nodes = {
      min_size     = 1
      max_size     = 3
      desired_size = 3
      instance_types = ["t3.micro"]
```

```
}

# 3. Security Group & EC2 for Jenkins
resource "aws_security_group" "jenkins_sg" {
  name        = "jenkins_sg"
  vpc_id      = module.vpc.vpc_id

  ingress {
    from_port   = 8080
    to_port     = 8080
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }
  ingress {
    from_port   = 22
    to_port     = 22
    protocol    = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }
  egress {
    from_port   = 0
    to_port     = 0
    protocol    = "-1"
    cidr_blocks = ["0.0.0.0/0"]
  }
}

resource "aws_instance" "jenkins_server" {
  ami           = "ami-0c7217cdde317cfec"
  instance_type = "t3.micro"
  key_name      = "jenkins-key"
  subnet_id     = module.vpc.public_subnets[0]
  vpc_security_group_ids = [aws_security_group.jenkins_sg.id]
  associate_public_ip_address = true

  tags = {
    Name = "Jenkins-Server"
  }
}
```

```
module.eks.aws_eks_cluster.this[0]: Still creating... [09m00s elapsed]
module.eks.aws_eks_cluster.this[0]: Still creating... [09m10s elapsed]
module.eks.aws_eks_cluster.this[0]: Still creating... [09m20s elapsed]
module.eks.aws_eks_cluster.this[0]: Still creating... [09m30s elapsed]
module.eks.aws_eks_cluster.this[0]: Still creating... [09m40s elapsed]
module.eks.aws_eks_cluster.this[0]: Still creating... [09m50s elapsed]
module.eks.aws_eks_cluster.this[0]: Still creating... [10m00s elapsed]
module.eks.aws_eks_cluster.this[0]: Creation complete after 10m5s [id=trend-app-cluster]
module.eks.data.tls_certificate.this[0]: Reading...
module.eks.time_sleep.this[0]: Creating...
module.eks.data.tls_certificate.this[0]: Read complete after 1s [id=b1b539daa206035ae3c3e28288b0681fa1b462f3]
module.eks.aws_iam_openid_connect_provider.oidc_provider[0]: Creating...
module.eks.aws_iam_openid_connect_provider.oidc_provider[0]: Creation complete after 3s [id=arn:aws:iam::839922332162:oidc-provider/oidc.eks.us-east-1.amazona
ws.com/id/2DED17D6312BACC13B3866308697D7E4]
module.eks.time_sleep.this[0]: Still creating... [00m10s elapsed]
module.eks.time_sleep.this[0]: Still creating... [00m20s elapsed]
module.eks.time_sleep.this[0]: Still creating... [00m30s elapsed]
module.eks.time_sleep.this[0]: Creation complete after 30s [id=2026-04-11T08:44:41Z]
module.eks.module.eks_managed_node_group["app_nodes"].module.user_data.null_resource.validate_cluster_service_cidr: Creating...
module.eks.module.eks_managed_node_group["app_nodes"].module.user_data.null_resource.validate_cluster_service_cidr: Creation complete after 0s [id=23265500362
52650686]
module.eks.module.eks_managed_node_group["app_nodes"].aws_launch_template.this[0]: Creating...
module.eks.module.eks_managed_node_group["app_nodes"].aws_launch_template.this[0]: Still creating... [00m10s elapsed]
module.eks.module.eks_managed_node_group["app_nodes"].aws_launch_template.this[0]: Creation complete after 15s [id=lt-07c001e082efc8562]
module.eks.module.eks_managed_node_group["app_nodes"].aws_eks_node_group.this[0]: Creating...
module.eks.module.eks_managed_node_group["app_nodes"].aws_eks_node_group.this[0]: Still creating... [00m10s elapsed]
module.eks.module.eks_managed_node_group["app_nodes"].aws_eks_node_group.this[0]: Still creating... [00m20s elapsed]
module.eks.module.eks_managed_node_group["app_nodes"].aws_eks_node_group.this[0]: Still creating... [00m30s elapsed]
module.eks.module.eks_managed_node_group["app_nodes"].aws_eks_node_group.this[0]: Still creating... [00m40s elapsed]
module.eks.module.eks_managed_node_group["app_nodes"].aws_eks_node_group.this[0]: Still creating... [00m50s elapsed]
module.eks.module.eks_managed_node_group["app_nodes"].aws_eks_node_group.this[0]: Still creating... [01m00s elapsed]
module.eks.module.eks_managed_node_group["app_nodes"].aws_eks_node_group.this[0]: Still creating... [01m10s elapsed]
module.eks.module.eks_managed_node_group["app_nodes"].aws_eks_node_group.this[0]: Still creating... [01m20s elapsed]
module.eks.module.eks_managed_node_group["app_nodes"].aws_eks_node_group.this[0]: Still creating... [01m30s elapsed]
module.eks.module.eks_managed_node_group["app_nodes"].aws_eks_node_group.this[0]: Still creating... [01m40s elapsed]
module.eks.module.eks_managed_node_group["app_nodes"].aws_eks_node_group.this[0]: Creation complete after 1m42s [id=trend-app-cluster:app_nodes-20260411084456
740100000014]

Apply complete! Resources: 54 added, 0 changed, 0 destroyed.
vicky@ubuntu:~/Trend$
```

Clusters (1) Info

Delete | Create cluster

Filter clusters

Cluster name	Status	Kubernetes version	Support period	Upgrade policy	Created	Provider
trend-app-cluster	Active	1.29	End of support	Extended support	28 minutes ago	EKS

Your VPCs (1) Info

Last updated less than a minute ago | Actions | Create VPC

Find VPCs by attribute or tag

Name = trend-app-vpc | Clear filters

Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv
trend-app-vpc	vpc-0a45d49f0b961f4cf	Available	–	–	Off	10.

Load balancers (1/1) What's new?

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Actions | Create load balancer

Filter load balancers

Name	Type	Scheme	IP address type	VPC ID	Availability Zones	Security groups	DNS
a8e97c6c648284fe28ea...	classic	-	-	vpc-099961d76e1221cb7	2 Availability Zones	sg-0ac37ea06d3959d9...	a

Load balancer: a8e97c6c648284fe28ea88911586e1bc

Details | Listeners | Network mapping | Security | Health checks | Target instances | Monitoring | Attributes | Tags

Details

Load balancer type
Classic

Scheme
Internet-facing

Status
2 of 2 instances in service

Hosted zone
Z35SXDOTRQ7X7K

VPC
vpc-099961d76e1221cb7

Availability Zones
subnet-0b31a5e9a08b9046e us-east-1b (use1-az2)
subnet-00f015917d40f37ef us-east-1a (use1-az1)

Date created
April 11, 2026, 19:16 (UTC+05:30)

Instances (4) Info

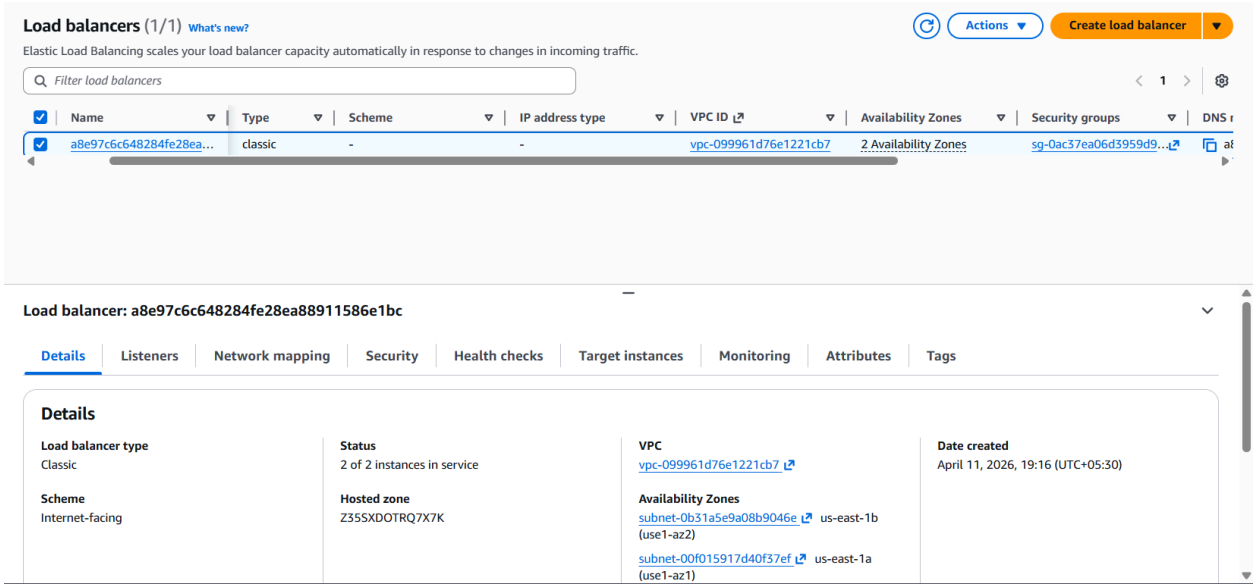
Connect | Instance state | Actions | Launch instances

Saved filter sets: Choose filter set

Find Instance by attribute or tag (case-sensitive)

Instance state = running | Clear filters

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
Jenkins-Server	i-0d463a9b57d5ed123	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1a	ec2-44-211-
app_nodes	i-0973c39e1cac84713	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1a	–
app_nodes	i-062e8287ab3018859	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1b	–



Continuous Integration & Deployment (CI/CD)

- Orchestrated a 4-stage Jenkins pipeline: Checkout, Docker Build, Docker Push, and EKS Deployment.
- Successfully executed Build #6, which automated the entire flow from code commit to cluster update.
- Pipeline stages:
  - Stage 1 — Clone Repository from GitHub
  - Stage 2 — Build Docker Image
  - Stage 3 — Push image to Docker Hub
  - Stage 4 — Deploy to EKS using kubectl

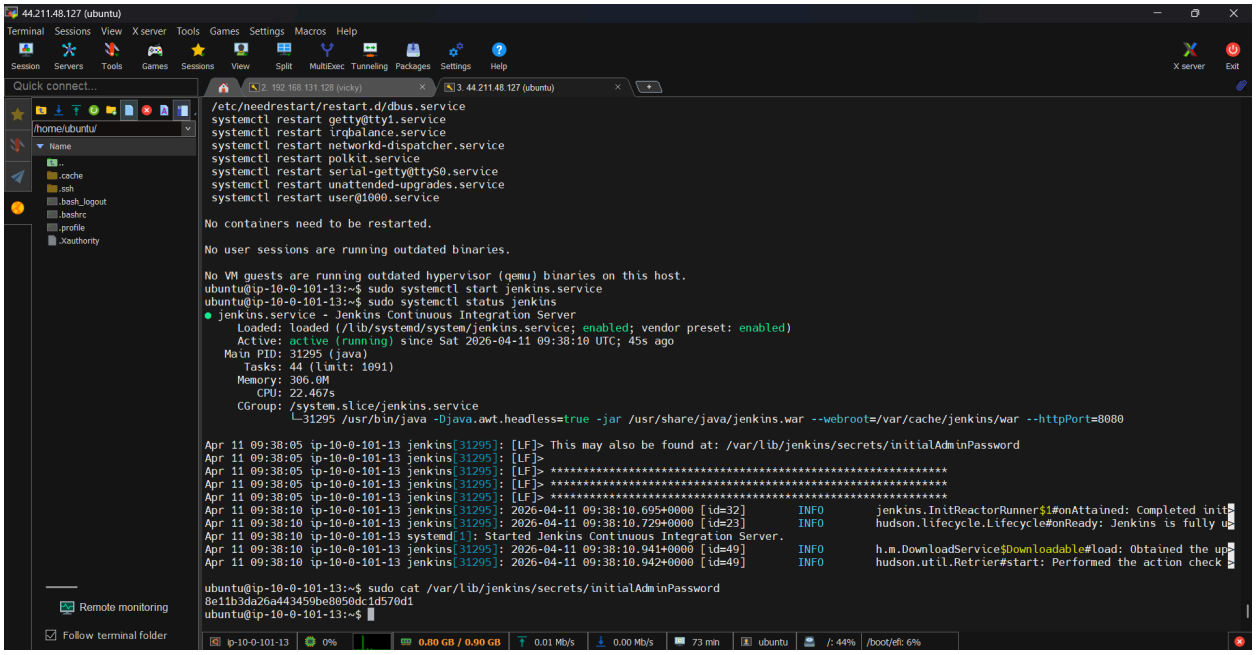
Configuration

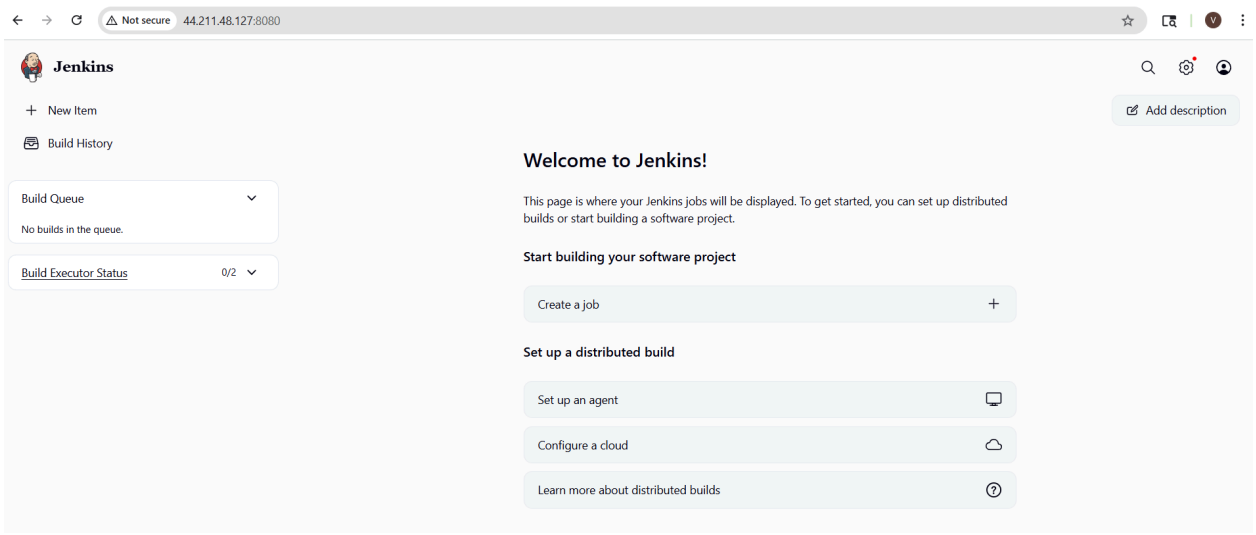
Installed Plugins
Docker, Docker Pipeline, Git, Pipeline, Kubernetes CLI

Configured Credentials
GitHub Token, Docker Hub Password

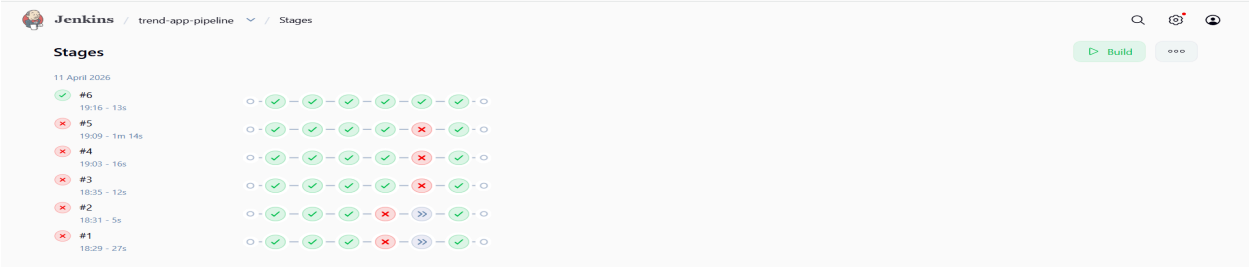
Automation Setup
GitHub webhook configured to automatically trigger builds on code push

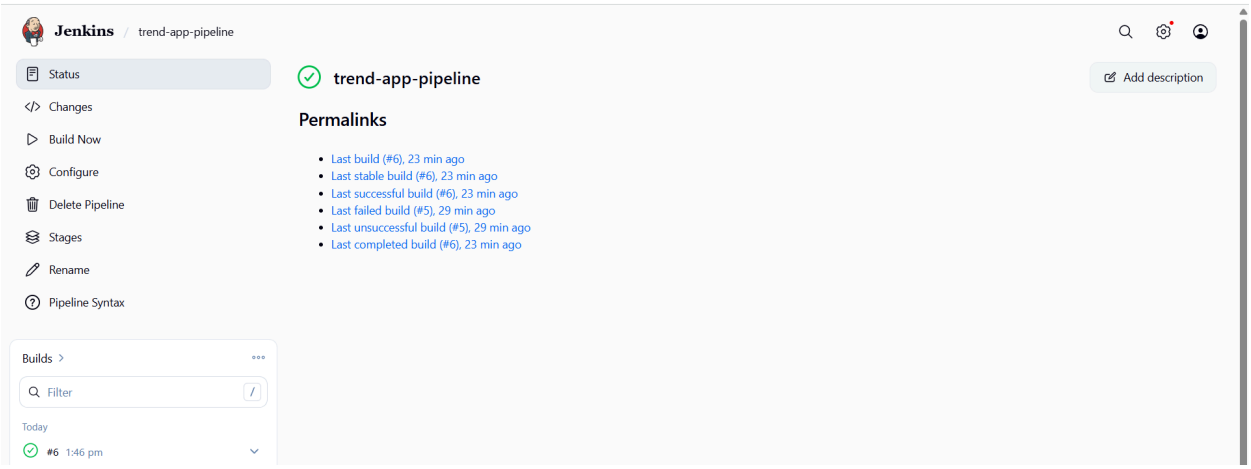
Access & Deployment Configuration
kubeconfig configured for cluster access, AWS credentials configured for Jenkins user


44.211.48.127 (ubuntu)
/etc/needrestart/restart.d/dbus.service
systemctl restart getty@tty1.service
systemctl restart irqbalance.service
systemctl restart networkd-dispatcher.service
systemctl restart polkit.service
systemctl restart serial-getty@ttyS0.service
systemctl restart unattended-upgrades.service
systemctl restart user@1000.service
No containers need to be restarted.
No user sessions are running outdated binaries.
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-10-0-101-13:~$ sudo systemctl start jenkins.service
ubuntu@ip-10-0-101-13:~$ sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
Loaded: loaded (/lib/systemd/system/jenkins.service; enabled; vendor preset: enabled)
Active: active (running) since Sat 2026-04-11 09:38:10 UTC; 45s ago
Main PID: 31295 (java)
Tasks: 44 (limit: 1091)
Memory: 306.0M
CPU: 22.467s
CGroup: /system.slice/jenkins.service
└31295 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080
Apr 11 09:38:05 ip-10-0-101-13 jenkins[31295]: [LF]> This may also be found at: /var/lib/jenkins/secrets/initialAdminPassword
Apr 11 09:38:05 ip-10-0-101-13 jenkins[31295]: [LF]>
Apr 11 09:38:05 ip-10-0-101-13 jenkins[31295]: [LF]> **********
Apr 11 09:38:05 ip-10-0-101-13 jenkins[31295]: [LF]> **********
Apr 11 09:38:05 ip-10-0-101-13 jenkins[31295]: [LF]> **********
Apr 11 09:38:10 ip-10-0-101-13 jenkins[31295]: 2026-04-11 09:38:10.695+0000 [id=32] INFO jenkins.InitReactorRunner$1#onAttained: Completed init
Apr 11 09:38:10 ip-10-0-101-13 jenkins[31295]: 2026-04-11 09:38:10.729+0000 [id=23] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is fully u
Apr 11 09:38:10 ip-10-0-101-13 systemd[1]: Started Jenkins Continuous Integration Server.
Apr 11 09:38:10 ip-10-0-101-13 jenkins[31295]: 2026-04-11 09:38:10.941+0000 [id=49] INFO h.m.DownloadService$Downloadable#load: Obtained the up
Apr 11 09:38:10 ip-10-0-101-13 jenkins[31295]: 2026-04-11 09:38:10.942+0000 [id=49] INFO hudson.util.Retrier#start: Performed the action check
ubuntu@ip-10-0-101-13:~$ sudo cat /var/lib/jenkins/secrets/initialAdminPassword
8e11b3da26a443459be8050dc1d570d1
ubuntu@ip-10-0-101-13:~$

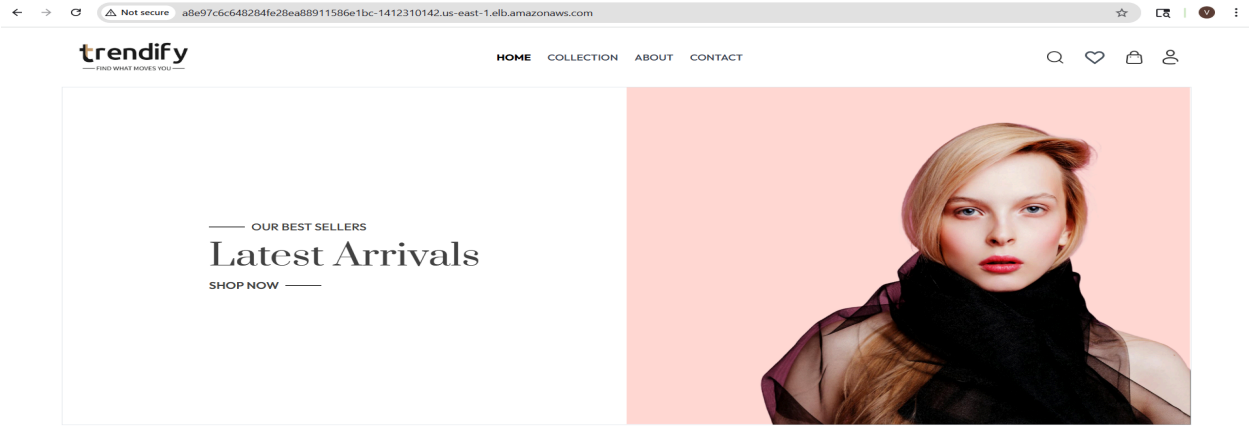

Not secure 44.211.48.127:8080
Jenkins
New Item
Build History
Build Queue
No builds in the queue.
Build Executor Status 0/2
Add description
Welcome to Jenkins!
This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.
Start building your software project
Create a job
Set up a distributed build
Set up an agent
Configure a cloud
Learn more about distributed builds

Not secure 44.211.48.127:8080/manage/credentials/store/system/domain/_/
Jenkins / Manage Jenkins / Credentials / System / Global
Global
Add Credentials
Credentials that should be available everywhere.
dockerhub-creds wikkics/******
kubeconfig-creds config.txt





Orchestration & Final Delivery

- Used `kubectl` to deploy Kubernetes manifests to the EKS worker nodes.
- Resolved resource scheduling (pending pods) by scaling the node group, resulting in a live, healthy deployment.

```
vicky@ubuntu:~/Trend$ kubectl get pods
NAME                          READY   STATUS    RESTARTS   AGE
trend-app-7c79ccd686-2mttz    1/1     Running   0          15m
vicky@ubuntu:~/Trend$
```